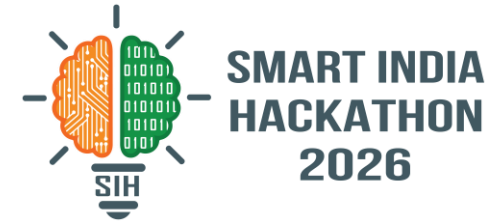


SMART INDIA HACKATHON 2026



MENTOR MATE

An AI Academic Learning Assistant Built on Real Student Data

Problem Statement Title: Autonomous Academic Learning Twin for Real-Time Personalized Learning

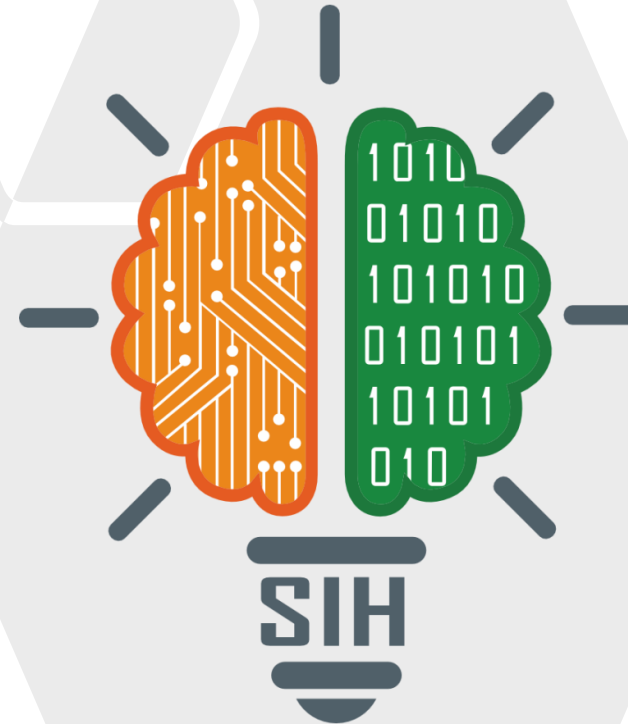
Problem Statement ID: SIH26207

Theme: Smart Education

Category: Software Edition

Team Name: Mentor Mate

Key Technologies Used: Next.js, React, FastAPI (Python), Vision AI (OCR), MongoDB



PROPOSED SOLUTION: MENTOR MATE

THE REAL PROBLEM

Same Recorded Videos for Everyone

Every student is given the exact same video playlist, ignoring what they already know and where they struggle.

Handwritten Notes Get Ignored

Students spend hours writing notes in school and college lectures, but current apps cannot read or use these notes.

Passive Watching Without Practice

Watching long videos gives a false feeling of learning. When exam day comes, students struggle to solve real questions.

OUR SOLUTION

Learns From Student Notes

Students simply take a picture of their notebook or upload lecture PDFs. The system reads the text and formulas directly.

Creates Instant Practice Tests

Instead of using old static questions, the system generates fresh practice questions based on what the student uploaded.

Guided Socratic Hints

When a student makes a mistake, the AI gives helpful hints step-by-step so the student understands how to solve it themselves.

KEY HIGHLIGHTS

No Fixed Question Banks

Questions are created fresh for the specific chapter or topic the student needs to practice.

Clear View of Weak Areas

Clearly identifies which topics had wrong answers, so students know exactly what needs revision.

Grounded in Real Student Input

Everything on the student's dashboard comes directly from their own notes, tests, and actual study history.

Frontend: Next.js & React (Clean student UI) | **Backend:** FastAPI / Python (Fast & lightweight API) | **AI & OCR:** Vision AI (Reads notes) + LLM (Generates tests) | **Database:** MongoDB (Stores notes & scores)

STEP 1: UPLOAD & READ NOTES**Upload Notebook Snaps & PDFs**

Students can upload photos of handwritten spiral notebooks, textbook pages, or lecture slides.

Extracts Text & Formulas

Vision AI reads the handwriting, extracts core formulas, and creates a clear summary of the page.

Quick Self-Check Drills

Instantly generates 2 to 3 self-test questions from the uploaded page to confirm the student understood it.

STEP 2: DYNAMIC PRACTICE TESTS**Choose Any Class Topic**

Students can test themselves either on their uploaded notes or type in any topic they learned in class today.

Generates 5 Practice MCQs

AI creates 5 fresh multiple-choice questions on the spot, complete with 4 clear options and worked solutions.

Immediate Answer Review

Shows student what they got right, what was wrong, and the step-by-step method to solve each question.

STEP 3: HINTS & WEAK SPOTS**Step-by-Step Guiding Hints**

Instead of giving direct answers, the AI tutor gives small hints so students think and solve it themselves.

Flags Topics for Revision

Keeps a list of questions and topics where the student made mistakes so they can review them before exams.

Free Course Recommendations

Connects weak topics directly to free, high-quality chapters from NCERT and NPTEL for extra practice.

HOW WE BUILD IT (FEASIBILITY)

Fast & Lightweight Backend

FastAPI handles note uploads and test requests asynchronously, keeping the app fast and responsive even with multiple users.

Zero Licensing Costs for Content

Uses verified, free open educational content (NCERT textbooks, NPTEL/SWAYAM), avoiding expensive content creation or licensing.

Low Server Storage Costs

We store text summaries, formulas, and test scores in MongoDB rather than heavy video files, keeping cloud costs minimal.

Tested Working Prototype

Core features like user authentication, note parsing, and dynamic question generation are tested with automated test scripts.

REAL CHALLENGES & OUR FIXES

Messy Handwriting & Poor Photos

Fix: Use high-accuracy Vision AI models and give students a quick screen to review or edit extracted text before generating tests.

Risk of AI Hallucinating Answers

Fix: Strictly ground test questions in the student's uploaded notes and standard syllabus topics with verifiable solutions.

AI API Call Costs

Fix: Cache common syllabus topics and use fast, cost-effective models for routine practice quizzes.

Keeping Study Routine Simple

Fix: Allow students to set simple daily goals (e.g. 1 to 2 hours) so they stay consistent without feeling overwhelmed.

TARGET AUDIENCE & PRACTICAL BENEFITS

SCHOOL (CLASS 9-10)

Focus Areas: 10th CBSE, ICSE, and State Board exams, plus basic science and math foundations.

How It Helps: Turns notebook pages into quick revision quizzes so students practice actively instead of just reading.

Student Benefit: Builds regular daily study habits and strengthens basic concepts before senior high school.

SENIOR SECONDARY (11-12)

Focus Areas: 12th Boards and entrance exams like JEE, NEET, and State Engineering CETs.

How It Helps: Provides regular MCQ problem-solving with step-by-step solutions to practice numericals and formulas.

Student Benefit: Helps students quickly spot which chapters they keep getting wrong so they can review them in time.

COLLEGE UNDERGRADS

Focus Areas: Semester university exams, GATE, and campus placement prep (DSA and core subjects).

How It Helps: Summarizes long professor slides and generates test drills for theoretical and programming concepts.

Student Benefit: Saves time organizing college study materials and provides quick self-testing before exams.

PRACTICAL BENEFITS OF MENTOR MATE

Practice From Your Own Notes: Students practice from what their teacher actually taught in class, without having to hunt for new questions.

Active Problem Solving: Taking short quizzes and getting stepped hints helps students remember concepts much better than passively watching videos.

Clear Revision Focus: Shows students clearly where they made mistakes, so they spend their study time on topics that actually need work.

RESEARCH FOUNDATIONS & REFERENCES

ACTIVE RECALL & PRACTICE

Research Reference

Dunlosky et al. (2013), 'Improving Students' Learning with Effective Learning Techniques', Psychological Science in the Public Interest.

Key Finding

Extensive educational research proves that practice testing and active recall create much stronger long-term retention than passively re-reading or re-watching lectures.

How We Use It

Mentor Mate immediately creates practice questions from uploaded notes so students actively test their memory right after studying.

GUIDED SOCRATIC HINTS

Research Reference

Chi et al. (2001), 'Learning from Human Tutoring', Cognitive Science & Bloom (1984), 'The 2 Sigma Problem'.

Key Finding

Students understand complex concepts much better when guided with progressive hints to discover answers themselves, rather than being given direct solutions.

How We Use It

Our AI Mentor breaks down difficult questions into 2 progressive hints, guiding the student to solve the problem on their own.

STANDARD CURRICULUM

Research Reference

NCERT National Curriculum Framework & NPTEL / SWAYAM Open Education Guidelines (Ministry of Education).

Key Finding

Grounding practice questions and revision topics in standardized national syllabi ensures test questions match actual exam difficulty and patterns.

How We Use It

Our topic suggestions and diagnostic tests follow official NCERT and university syllabus structures, linking weak areas to free open courseware.